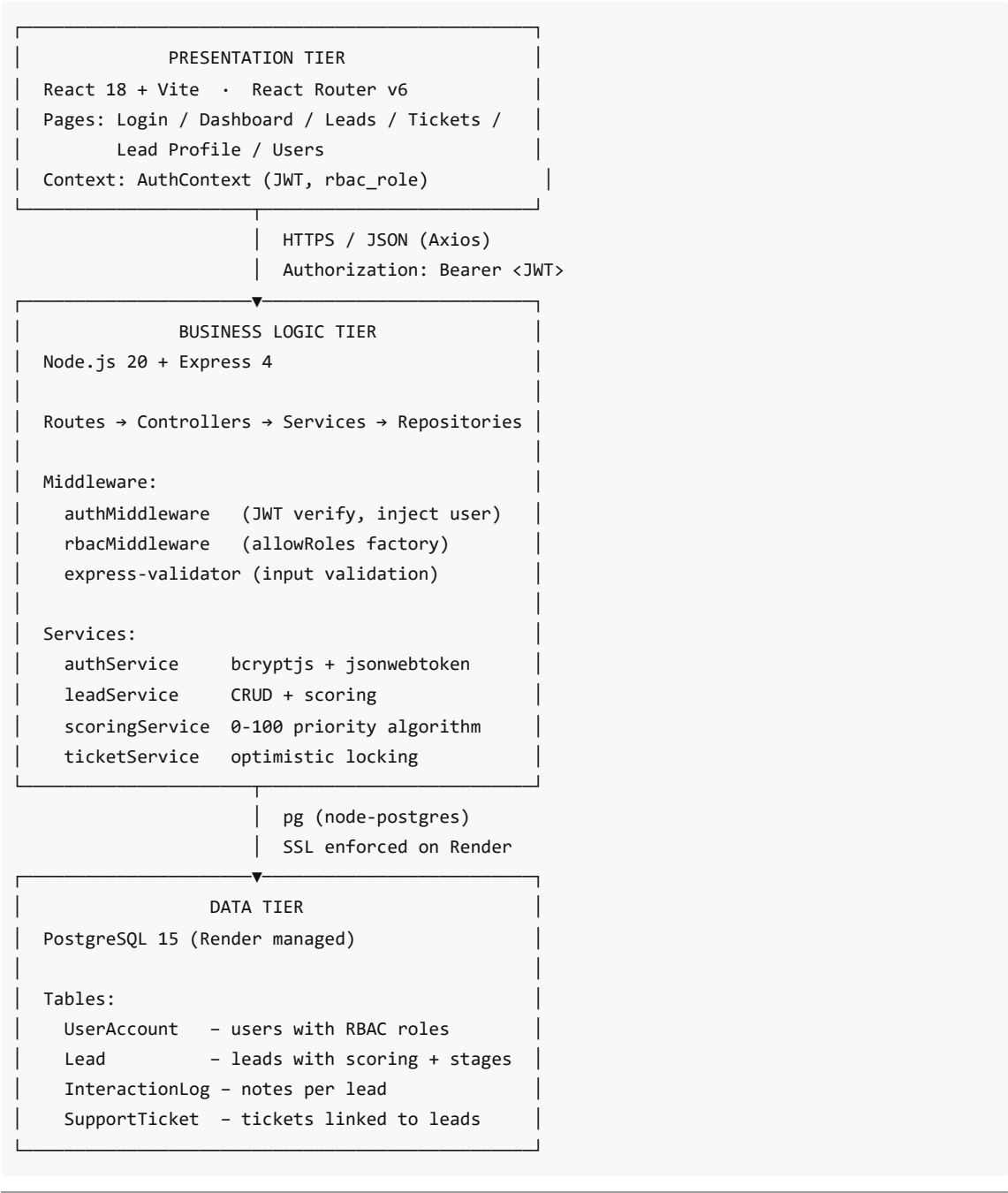


Architecture Description – WBOCRM

Overview

WBOCRM follows a classic **3-tier web application architecture** with a clear separation between presentation, business logic, and data layers. The system is designed for deployment on Render (cloud PaaS) with a PostgreSQL managed database.



Frontend Architecture

Layer	Technology	Purpose
-------	------------	---------

Build	Vite 5	Fast HMR dev server, optimised production bundle
UI	React 18	Component-based SPA
Routing	React Router v6	Client-side routing with protected routes
Auth state	React Context API	JWT + user object available app-wide
HTTP	Axios	Centralised API client with auth header injection
Deployment	Render Static Site	Served as pre-built static files with SPA rewrite

Route Protection

- ProtectedLayout — redirects to /login if no user in context
- RoleRoute — redirects to /dashboard if user.rbac_role not in allowed list

Backend Architecture

Request Lifecycle

```
HTTP Request
  → Express Router (/api/*)
    → authMiddleware (verify JWT, attach req.user)
      → rbacMiddleware (allowRoles check)
        → Controller (parse request, call service)
          → Service (business rules, orchestration)
            → Repository (parameterised SQL queries)
              → PostgreSQL pool
```

Controller → Service → Repository Pattern

Each domain (leads, tickets, users, auth) has three files:

- Controller:** HTTP concerns only — parse body, call service, return status codes
- Service:** Business logic — validation, scoring, conflict detection
- Repository:** SQL only — all queries in one place, no business logic

Key Design Decisions

JWT Authentication

Tokens are signed with HS256, expire in 30 minutes, and carry user_id and rbac_role. The secret is injected via JWT_SECRET environment variable.

RBAC Middleware

allowRoles(...roles) returns an Express middleware that reads req.user.rbac_role and returns 403 if the role is not in the allowed list. Roles: admin, sales, support.

Lead Scoring Algorithm

Calculated at creation time. Five weighted metrics (max 100 pts):

- Calls completed: up to 25 pts (calls / 20 × 25)
- Meetings held: up to 25 pts (meetings / 5 × 25)
- Budget potential: up to 20 pts (budget / 100 × 20)

- Company size: 5 / 10 / 20 pts (small / medium / enterprise)
- Email opens: up to 10 pts (email10pens / 20 × 10)

Optimistic Locking on Tickets

PUT /api/tickets/:id requires updated_at in the request body. If the DB record's updated_at differs, a 409 CONFLICT is returned, preventing lost updates (NFR-ST-05).

Dashboard Auto-Refresh

DashboardPage polls /api/dashboard every 2 seconds via setInterval with cleanup on unmount (NFR-ST-12).

Database Schema

```
UserAccount (user_id PK, user_email UNIQUE, user_password, rbac_role, full_name, created_at)
Lead (lead_id PK, email UNIQUE, contact_name, priority_score, pipeline_stage,
      deal_value, campaign_id, user_id FK→UserAccount, created_at)
InteractionLog (log_id PK, note_text, timestamp, lead_id FK→Lead, user_id FK→UserAccount)
SupportTicket (ticket_id PK, description, status, priority_level, updated_at,
               lead_id FK→Lead, assigned_to FK→UserAccount)
```

Foreign Key Cascade Strategy

- Lead deleted → InteractionLog and SupportTicket rows deleted (application-level, in order)
- UserAccount deleted → Lead.user_id set to NULL (no orphan leads)

Security Considerations

- Passwords hashed with bcryptjs, saltRounds=12
- JWT secret loaded from environment, never hard-coded
- All SQL uses parameterised queries (no string interpolation) — prevents SQL injection
- CORS restricted to FRONTEND_URL environment variable
- Input validation via express-validator on auth routes
- SSL enforced for all DB connections in production